

Two-stage validation plan for any downscaling moisture model, framed around full use of dense point datasets.

The dense spatial dataset allows sub-grid validation of the downscaling itself, not just station-level validation. Comparable studies often validate against sparse in-situ stations; this adds a “microscope” over within-site terrain redistribution as well as seasonal dynamics.

Before a new validation run, pull the latest DownscalingMoistureModel process-model branch from <https://github.com/thestochasticman/DownscalingMoistureModel/tree/process-model> at the local Volumes/Dmitry_work/borevitz_projects/DownscalingMoistureModel location.

Full use of the dense spatial point datasets

There are three datasets

1. Esdale, 540 points, 0.2pts/ha, autumn/winter of 2025 only

This site represents the most spatial/terrain type coverage, so terrain residual inference is most valuable here.

2. Tarrawarra, 312pts/ha, 50ha only, full year 1995/96. No SMIPS available.

This site represents the densest spatial sampling, so terrain residual mapping is a necessary output.

3. Llara rehydration projects. 0.4pts/ha, multiple years in 2021 to 2024, 32 points in two separate 40ha paddocks.

This site represents the greatest temporal coverage, so seasonal residual inference is most valuable here.

The datasets should be used in three complementary ways:

Stage 1: Validation target

- Use all point/date observations as independent external validation.
- Use dense spatial coverage to map where residuals cluster in space and time

Stage 2: Local calibration experiment

- Use controlled subsets of the same dense dataset as “spikes” while holding out the rest for evaluation. Determine the minimum amount of points and best calibration procedure to reach average NSE > 0.3 per site.

The most important design rule: keep calibration and validation separated by point, date, or terrain block so the experiment measures genuine transfer, not memorisation. Treat only spatial + temporal blocking as the true measure of model performance.

A strong final analysis would produce:

- Summary error metrics overall, and stratified by terrain (only most notable) and seasonality/wetness state (only most notable)
- maps of prediction quality as interpolated surfaces
- maps of residual bias as interpolated surfaces
- model-input-space plots showing where errors occur
- learning curves showing accuracy gains from local spiking
- recommendations for the minimum local sampling density needed for reliable local prediction, in particular **reaching average NSE of >0.4**
- identification of which terrain types most need local calibration
- A final summary report showing only the most notable results as well as any notes necessary for inference of results.

Stage 1 — Independent validation on unseen high-density point data

Objective: test how well the DMMs transfer to an unseen, densely sampled site, without using those points for training. The key is to make the validation protocol operate on a common prediction table, not on the internals of the RF or process models. That way both models are judged only by: “given this point, date, depth, and observed soil moisture, what did the model predict?”

Using a dense dataset is desirable because the model is intended to capture ‘paddock’ scale variation in soil moisture. Using SM stations separated by kilometers is okay for training, but validation must capture within-site bias/variation.

Validation pipeline

1. Ingest dense point dataset that contains rows of individual points
 - Point_id
 - SM%
 - Date
 - Lat
 - Lon
2. Generate model predictions for every sampling date in the point dataset and cache locally.
3. Extract predicted soil moisture at every point/date.
4. Create a model-agnostic validation table that specifies
 - Model_name (e.g. RF_gradientboosting, Process_model)
 - Point_id
 - Pred_sm
 - Obs_sm
 - date,
 - season,
 - terrain strata,

- wetness phase

Not necessarily model parameters but validation covariates to determine systematic spatial and temporal over/under prediction.

5. Compare predictions against observed point soil moisture using these metrics:
 - NSE / R^2
 - Pearson r
 - RMSE
 - ubRMSE
 - bias
 - per-point and pooled summaries
6. Map prediction quality spatially, especially per-point NSE/ R^2 , bias and RMSE as interpolated surfaces.

The key analysis is: are bad predictions clustered in particular terrain zones and/or seasons? For example, high TWI/low slope zones, high-HLI exposed slopes, drainage accumulation zones, unusually sandy/clayey pixels, summer vs. winter.

Seasonal/wetness state bias should also be examined.

seasonal mean bias: $\text{mean}(\text{pred} - \text{obs})$ by season
 dry-condition bias: bias when observed moisture is in the driest 25%
 wet-condition bias: bias when observed moisture is in the wettest 25%

The last two are important because a model may look okay overall but systematically overpredict dry soils or underpredict wet soils. This is very relevant for paddock-scale management.

Produce a report that shows model error metrics, observed vs. predicted timeseries, output SM prediction maps of the wettest and driest date, and any notable stratified terrain/season error differences between the two models.

Stage 2 — Sensitivity to local training-data “spiking”

Objective: quantify how much predictive skill improves when increasing amounts of local data are added to the training process. Explore different local calibration ‘adaptation’ techniques like constant residual offset, seasonal offset, affine correction, etc... The key is the best combination of minimum points + calibration technique to reach an average NSE > 0.4.

Read Volumes/Dmitry_work/borevitz_projects/DMM_validation/docs/local_calibration_layers.md for more info on exact techniques.

The core experiment is a learning curve and a comparison of local calibration techniques:

OzNet-trained model only
OzNet + 1 local point
OzNet + 3 local points
OzNet + 5 local points
OzNet + 10 local points
OzNet + 25% local points
OzNet + 50% local points
OzNet + all available calibration points

For each spike level, evaluate on held-out local points that were not used in training. A main consideration has become that local calibration has to work under the philosophy that any new unseen sites will have no soil moisture information, so local 'spiking' has no information to work with to inform where model performance is bad. I think the most defensible approach is to do an almost random point selection, maybe prioritising areas where the global model is known to fail (like very dry or very wet, landowners know this information intuitively, and our previous analysis demonstrates clear compression problems) and seeing the minimum amount of points to reach $NSE > 0.4$.

For the top three best experiments, report each as a bar graph comparison of 'global' vs. 'calibrated' change

- $\Delta NSE / \Delta R^2$ relative to the unspiked model
- $\Delta RMSE$
- ΔubRMSE
- Δbias
- number of local points required to reach $NSE > 0.4$
- number of local observations required to halve bias
- Which model responds best to spiking

Final output figures for each site should also include

- new calibrated observed vs predicted timeseries
- output maps of wettest + driest compared to uncalibrated models

This stage is especially novel if framed as "how much local calibration is needed to make national-scale downscaling locally reliable?" Comparable studies often evaluate model transfer or retrain with local data, but fewer explicitly quantify the marginal value of dense local observations as a function of sampling density and terrain representativeness.

Write up a report structured around site, i.e. Esdale, Tarrawarra, and Llara, with which model and calibration strategy performed best at each site, how each model responded to local spiking at each site, include the figures with inference notes, and a final summary section. Make it presentation ready.

